ADEPT — TCN Base Learner

Phase 3 Results: Analysis and Next Steps

Guilherme Leitão

Master Thesis — IST / INESC-ID

May 11, 2026

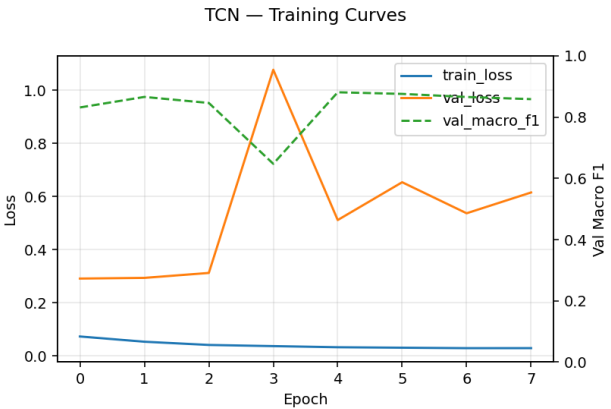
What was trained

- **Task:** same binary classification as RF Phase 2 — 5 benign + 6 adversarial sources (including the new adversarial-only `ddos_kitsune`), shared flow-atomic 70/15/15 split (seed=42), same exclusions (`portscan_adv_*` empty; `*_kitsune` variants for brute-force / portscan defensively excluded).
- **Input:** `sequence.npy` per source — each sample is a $[T{=}10,\ F{=}1090]$ window (1 088 nPrint bits + IPAT z-score + Direction); lazy memmaps.
- **Architecture:** 3-block causal TCN, channels $[128, 128, 128]$, kernel 3, dilations $\{1, 2, 4\}$, RF $= 29 \gg T$, mean+max pooling $\rightarrow$ linear. **806 018 parameters.** AMP enabled (CUDA).
- **Training:** AdamW ($\eta{=}10^{-3}$, wd$= 10^{-4}$), cosine LR, `class_weight=balanced`, early-stop patience 3.

Headline metrics (test, overall) — Δ vs. previous run

	Macro F1	ROC AUC	PR AUC	Recall (adv)	Precision (adv)
Current	0.883	0.938	0.840	0.861	0.761
Previous	0.585	0.707	0.169	0.185	0.266

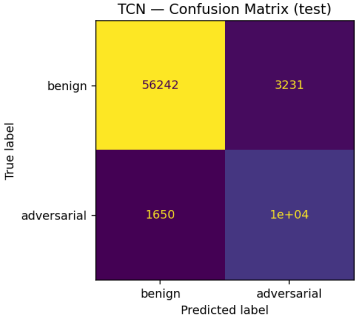
Training behaviour



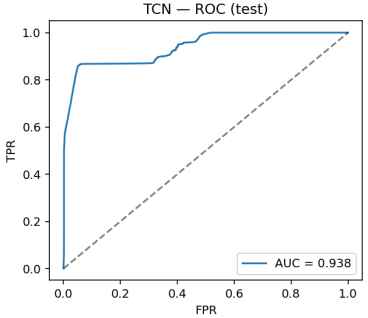
Ep.	Tr. loss	Val. loss	Val. F1
0	0.073	0.291	0.831
1	0.053	0.293	0.866
2	0.041	0.312	0.846
3	0.036	**1.077**	**0.647**
4	**0.032**	0.511	**0.881**
5	0.030	0.654	0.876
6	0.029	0.537	0.866
7	0.029	0.615	0.859

- **Best checkpoint: epoch 4** (val F1 = 0.881). Val macro-F1 climbs 0.83 → 0.88 with a single dip at epoch 3 (val loss to 1.08, F1 down to 0.65); recovery at epoch 4 is decisive rather than oscillating.
- Early stopping (patience=3) fires after epoch 7.
- Training: 471.6 s on CUDA + AMP ($\approx$ 59 s/epoch, $\sim 4.5\times$ previous run due to larger adv. pool).

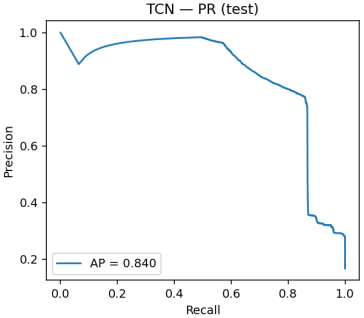
Confusion matrix & curves



Confusion Matrix



ROC (AUC = 0.938)



Precision–Recall (AP = 0.840)

- **Adversarial recall = 86.1%**: 4.6× jump from 18.5% — the model now flags most adversarial windows.
- **PR-AUC = 0.840** (was 0.169), **ROC-AUC = 0.938** (was 0.707): real probabilistic discrimination.
- Driver: the `ddos_kitsune` reinforcement gave the TCN enough gradient signal on the adversarial class to escape the previous benign-dominated minimum.

Per-variant performance (vs. balanced benign)

Variant	Test n	$F1_{adv}$	Recall	ROC AUC
brute_force_adv_both	32	0.694	0.781	0.971
brute_force_adv_iat	32	**0.438**	**0.438**	0.843
brute_force_adv_size	32	0.647	0.688	0.955
ddos_adv_both	40	**0.777**	1.000	1.000
ddos_adv_iat	40	**0.808**	1.000	0.978
ddos_adv_size	40	**0.800**	1.000	1.000
ddos_kitsune_adv_both	2,917	**0.782**	1.000	0.999
ddos_kitsune_adv_iat	2,917	**0.786**	1.000	0.972
ddos_kitsune_adv_size	2,917	**0.784**	1.000	0.999
mirai_adv_both / iat	841	**0.231 / 0.232**	0.202	0.70 / 0.68
mirai_adv_size	841	**0.607**	0.675	0.847
ssdp_flood_adv_both/iat/size	141	0.768 / 0.790 / 0.797	1.000	0.994–0.999

- **DDoS IAT: F1** 0.105 → 0.808 — the previous run's complete IAT blindness is gone.
- ddos_kitsune: F1 ≈ 0.78 across all 3 modes with perfect recall — the new source is uniformly well-detected.
- **Mirai partly recovered**: size mode F1 = 0.607 (was 0.000); both and iat still weak at ≈ 0.23 but now non-zero (was 0.000).
- **Brute force IAT regressed** (F1 0.775 → 0.438) — low-support, the bigger training pool pulled the decision boundary away from this rare class.

Where the TCN still struggles: Mirai & low-support brute-force

Mirai: partial recovery, not full

The previous run scored F1 = 0.000 on every Mirai mode — the model classified every Mirai adversarial window as benign. The augmented training pool flips this:

Mode	$F1_{adv}$ (was)
`mirai_adv_both`	**0.231** (0.000)
`mirai_adv_iat`	**0.232** (0.000)
`mirai_adv_size`	**0.607** (0.000)

Mirai's headers are highly uniform within the capture, so the temporal filters still latch onto the unchanged 1 088 nPrint bits and only weakly use the single IPAT z-score. `size` mode fares best because payload-length bytes *are* encoded in nPrint.

Feature geometry (unchanged from prior analysis)

- 1 088 nPrint header bits + 1 IPAT z-score + 1 Direction
- IPAT share = $1/1090 \approx$ **0.09%** vs. **16.7%** in the RF aggregates.
- `_iat` variants change exactly 1 of 1 090 features per packet; the conv filters learn to ignore it.

Brute-force IAT regression

Brute-force IAT F1 dropped from 0.775 to 0.438. With only 32 test windows the variance is large, but the trend is consistent: the model re-allocates capacity toward the dominant adversarial classes (`ddos_kitsune`, Mirai) at the expense of this small low-support class.

Contrast with RF: 16.7% IPAT feature share and aggregate mean/std/min/max recover Mirai cleanly (RF F1 $\geq$ 0.89 vs. TCN 0.23–0.61).

RF vs. TCN: head-to-head ($F1_{adv}$, balanced benign)

Variant	RF	TCN	Advantage
`brute_force_adv_both / iat / size`	**0.985 / 0.839 / 0.885**	0.694 / 0.438 / 0.647	**RF (+0.24 to +0.40)**
`ddos_adv_both / iat / size`	**0.952 / 0.864 / 0.976**	0.777 / 0.808 / 0.800	**RF (+0.06 to +0.18)**
`ddos_kitsune_adv_both / iat / size`	**0.955 / 0.951 / 0.960**	0.782 / 0.786 / 0.784	**RF (+0.17 to +0.18)**
`mirai_adv_both / iat / size`	**0.912 / 0.895 / 0.909**	**0.231 / 0.232 / 0.607**	**RF (+0.30 to +0.68)**
`ssdp_flood_adv_both / iat / size`	**0.949 / 0.956 / 0.953**	0.768 / 0.790 / 0.797	**RF (+0.16 to +0.18)**
Overall macro-F1	**97.94%**	**88.31%**	**RF (+9.6 pts)**

- **The TCN is now a credible base learner**: macro-F1 went from 0.585 to 0.883 and it tracks the RF within ∼10 pts on every variant except Mirai `both/iat`.
- **Gap collapsed from +39.3 pts to +9.6 pts**: the `ddos_kitsune` reinforcement made the temporal head trainable on adversarial structure.
- **RF still wins on every variant**: the aggregate-statistics view continues to give the RF a structural edge — especially on Mirai `both/iat` (+0.68 pts).

Conclusions

1. **Overall macro-F1 = 88.31%, ROC-AUC = 0.938, PR-AUC = 0.840** — the TCN crossed from *not working* (prev. 0.585 / 0.707 / 0.169) to a competitive base learner. Adding `ddos_kitsune` was the decisive change.
2. **DDoS IAT recovered**: F1 went from 0.105 to 0.808 — IAT-only perturbations are no longer invisible to the temporal head.
3. **Mirai `both/iat` are the remaining weak spot** (F1 $\approx$ 0.23, up from 0.000): partial recovery, but the structural symmetry between benign and adversarial Mirai windows still defeats the bit-level representation. `mirai_adv_size` reached F1 = 0.607 because payload-length bytes show up in nPrint.
4. **Brute-force IAT regressed** (0.775 $\rightarrow$ 0.438) — low-support side-effect of re-allocating capacity toward the dominant adversarial classes.
5. **RF and TCN are now genuinely complementary**: gap collapsed from 39 to 10 pts, but the RF still leads on every variant. Phase 4 meta-learner motivation is preserved (route Mirai/IAT to RF; both contribute on `ddos_kitsune`, SSDP, DDoS).
6. **Open work**: IPAT reweighting and the N=20 sweep are still on the roadmap to close the Mirai `both/iat` gap.

Next steps

Short-term — fix TCN and build the ensemble

1. **IPAT reweighting**: augment the IPAT channel with a learned scalar or multi-scale temporal difference features to recover IAT-mode sensitivity without inflating the input dimension.
2. **Window size sweep**: evaluate $N \in \{10, 20\}$; longer windows provide more IAT contrast for slow-jitter attacks.
3. **Phase 4 — Mini-Transformer meta-learner** fusing RF + TCN logit vectors via self-attention. Expected to recover Mirai / IAT variants by routing to the RF when TCN probability is near-uniform.

Long-term — defense layer & write-up

4. **Phase 5 — Purification Transformer** (decoder-only Seq2Seq) projecting adversarial windows back onto the benign manifold; KV-caching for latency.
5. **Phase 6 — RL Adaptation Agent** re-weighting the ensemble online from TANTGen feedback + offline DPI validation.
6. **Dissertation write-up.**

Reproducibility

Artifacts under `ADEPT/data/models/tcn/`

- `tcn.pt` — best model state_dict (epoch 4)
- `metrics.json` — overall + per-source (`vs_all` / `vs_balanced` / `vs_same_base`) + per-label_str
- `split.npz`, `dataset_index.npz` — exact train/val/test partition, shared element-wise with RF via `assemble_index` (`seed=42`)
- `training_info.json`, `training_curves.json`, `tcn_config.json` — full training trace and config
- `*.png` — confusion matrix, ROC, PR, training curves

Re-run from scratch

```
python3 scripts/packet_to_tensor/process_all_attacks.py --tantgen
python3 scripts/tcn/run_tcn.py
```